WASTELAND PANTRY

SDG 12

Responsible Consumption and Production.

An AI/ML system that predicts how much food a planned meal will waste, so kitchens can cook just enough.

A weather forecast for your kitchen's food waste.

Presented by **Ashish Jakhmola**

Internship ID **IBMUEML005287**

Machine learning project
Python, pandas, scikit-learn

Aligned with UN SDG 12

Prototype trained on synthetic data

THE PROBLEM

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Kitchens plan by experience and guesswork,
but demand keeps changing.

Over-cooked food means
wasted money, ingredients and effort.

Waste is recorded after it happens,
so it doesn’t inform the next meal.

Decisions still rely on
paper logs, spreadsheets and gut feeling.

THE SOLUTION AND THE IDEA IN USE

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PLAN TODAY

INPUT

Manager enters tomorrow's meal plan and key details.

PREDICT

OUTPUT

Wasteland Pantry forecasts food waste in kilograms.

TAKE ACTION

DECIDE

Review prediction and adjust meals or portions before cooking.

LEARN AND IMPROVE

Kitchen records from real meals refine the model over time for better predictions.

WHO IT IS FOR

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BUILT FOR

Kitchens that already keep daily meal records.

PRIMARY USERS

College canteens and hostel mess committees.

ALSO FOR

Office and corporate cafeterias, community kitchens and NGOs.

EARLY ADOPTERS

Hostel messes and campuses with daily meal records.

BENEFITS

- Lower food cost
- Better planning
- Less wasted food for the environment

THE DATASET

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Synthetic dataset of

1,800

meal records and 18 columns

Covering

2 Jan 2023 to 30 Dec 2025

Target variable

Food_wasted/kg)

Inputs used (14)

weekday, meal time, menu category, event type, holiday flag, planned attendance rate, meals planned, expected meals, menu popularity score, portion size, forecast temperature, forecast rainfall, prior similar wasted meal, month

Left out on purpose

Record_id, raw Date, Actual_MealServed and Stockout_flag
only known after the meal, so using them would be data leakage

Data checks

no duplicates, no negative waste
21 missing temperature and
27 missing rainfall values
filled with the median

#	Column	Non-Null Count
0	Record_id	1800 non-null
1	Date	1800 non-null
2	Week_day	1800 non-null
3	Meal_Time	1800 non-null
4	Menu_category	1800 non-null
5	Event_type	1800 non-null
6	Holiday_flag	1800 non-null
7	Planned_Attendance_rate	1800 non-null
8	Meals_planned	1800 non-null
9	Expected_meals	1800 non-null
10	Menu_Popularity_score	1800 non-null
11	Portion_size(g)	1800 non-null
12	Forecast_temperature(C)	1779 non-null
13	forecast_rainfall(mm)	1773 non-null
14	Prior_similar_wasted_meal(kg)	1800 non-null
15	Actual_MealServed	1800 non-null
16	Stockout_flag	1800 non-null
17	Food_wasted(kg)	1800 non-null

METHODOLOGY

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1 → 2 → 3 → 4 → 5 → 6 → 7

1 DATA UNDERSTANDING

Run data checks and explore patterns in the meals data.

2 SELECT INPUTS WISELY

Choose relevant features and prevent data leakage.

3 CHRONOLOGICAL SPLIT

Train on 1,438 meals (to 23 May 2025)

Test on 362 meals (25 May to 30 Dec 2025)

4 PREPROCESS IN A PIPELINE

Fill missing numbers with medians.

One-hot encode text columns.

5 TRAIN BASELINES

Build two baselines:

Linear Regression

Random Forest

6 EVALUATE & ANALYZE

Measure with MAE, RMSE and R squared.

Assess errors and review important features.

7 SAVE & PREDICT

Save the model with joblib and generate predictions for a new meal.

results

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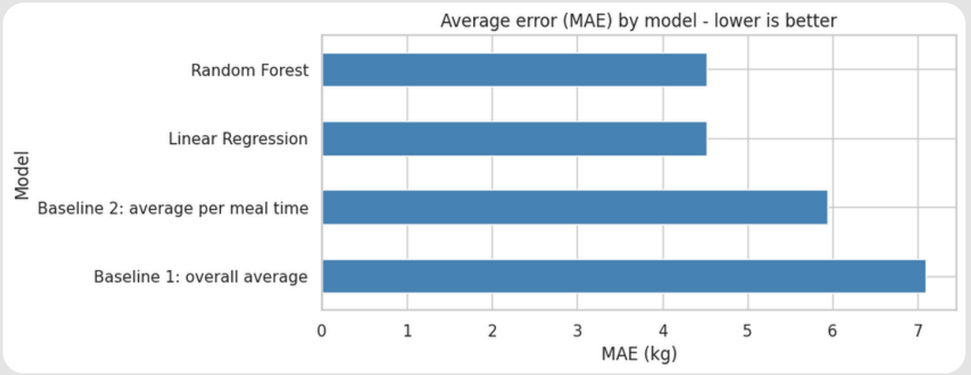
Models vs baselines on the 362 future test meals.

MODEL	MAE (KG)	RMSE (KG)	R2
Baseline 1 overall average	7.10	8.91	-0.001
Baseline 2 average per meal time	5.94	7.62	0.267
Linear Regression	4.52	5.81	0.574
Random Forest	4.52	5.80	0.575

	Model	MAE (kg)	RMSE (kg)	R2
0	Random Forest (with Prior column)	4.52	5.80	0.575
1	Random Forest (without Prior column)	4.49	5.77	0.579



Both models beat both baselines.

Linear Regression and Random Forest perform almost the same, so the simpler model works just as well on this data.

ERROR ANALYSIS AND FEATURE IMPORTANCE

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MODEL ERROR AT A GLANCE

average error by meal time

3.29 kg	4.46 kg	5.12 kg
Breakfast	Dinner	Lunch

average error by event

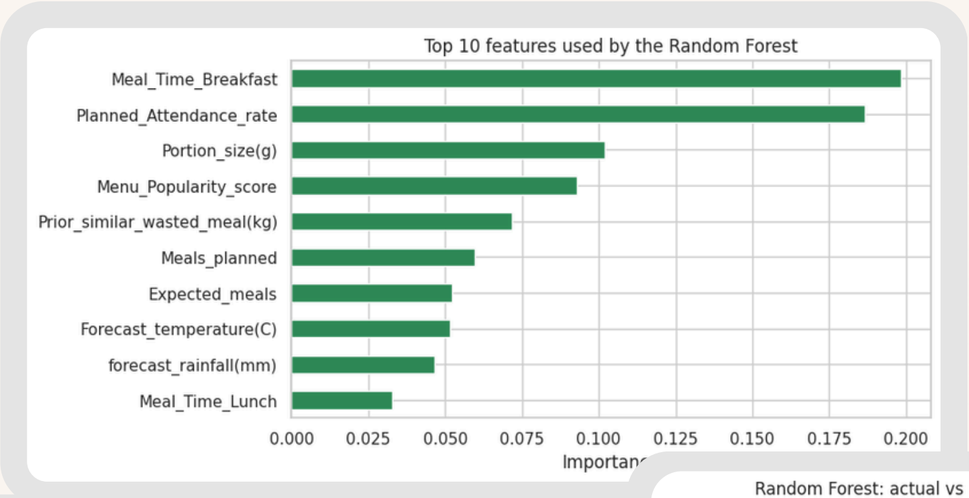
2.10 kg	5.19 kg	5.26 kg
Exam Period (lowest)	Festival	College Event

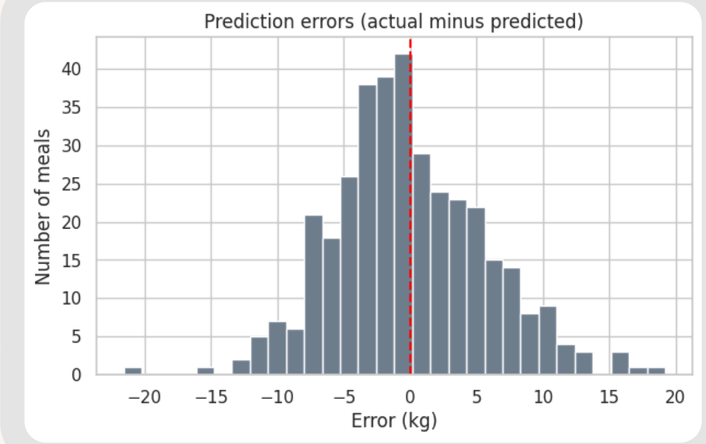
most used inputs

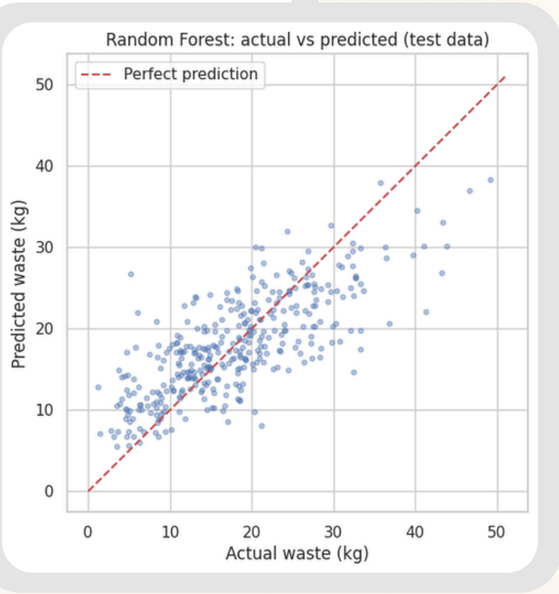
breakfast meal time
planned attendance rate
portion size
menu popularity

sensitivity check

removing the prior similar wasted meal column barely changed the score **(MAE 4.49 kg)**







LEAN CANVAS AND BUSINESS VIEW

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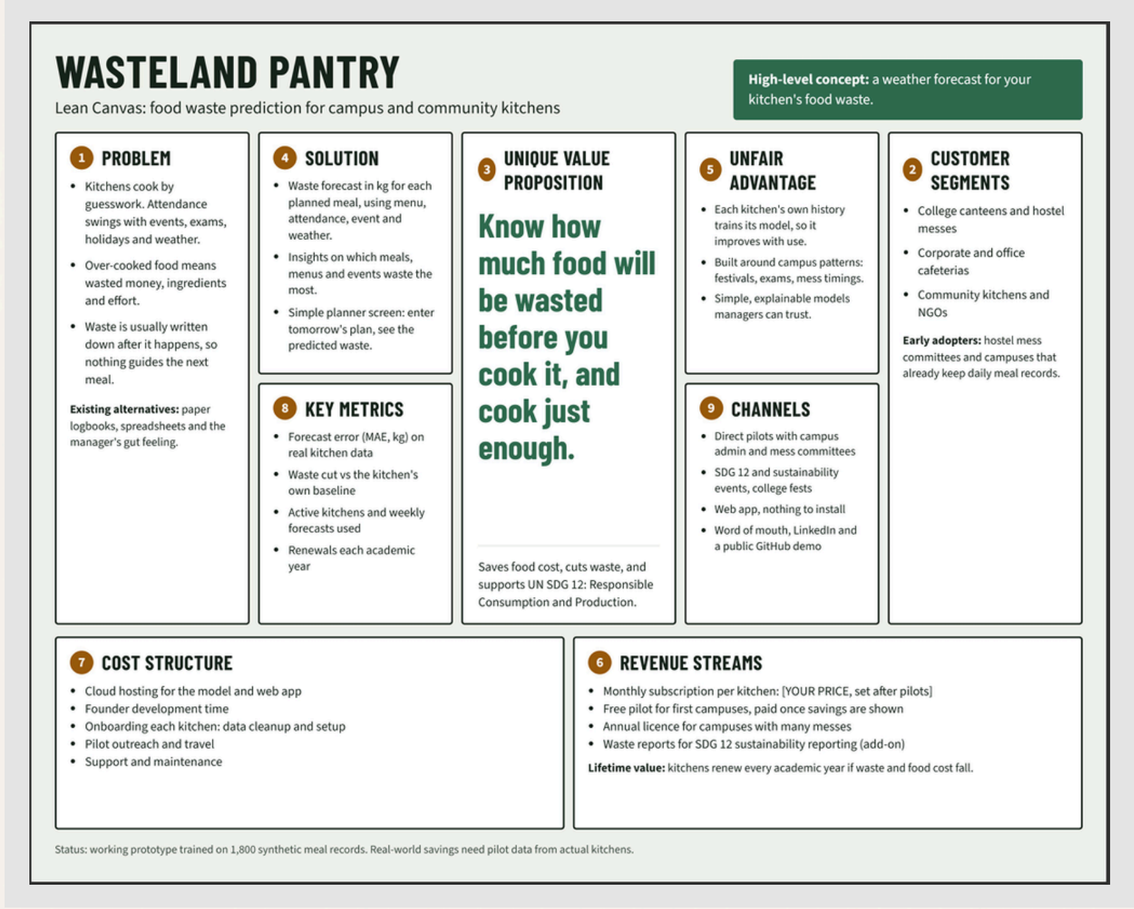
UNIQUE VALUE	know how much food will be wasted before you cook it, and cook just enough
CUSTOMERS	college canteens and hostel messes, corporate cafeterias, community kitchens and NGOs
CHANNELS	pilots with campus admin and mess committees, SDG 12 events and college festivals, web app, public GitHub demo
REVENUE IDEAS	monthly subscription per kitchen after free pilots, annual campus licence, SDG 12 waste reports add-on
KEY METRICS	forecast error (MAE in kg) on real data, waste cut vs the kitchen's own baseline, active kitchens, renewals



WASTELAND PANTRY

Lean Canvas: food waste prediction for campus and community kitchens

High-level concept: a weather forecast for your kitchen's food waste.

1 PROBLEM

- Kitchens cook by guesswork. Attendance swings with events, exams, holidays and weather.
- Over-cooked food means wasted money, ingredients and effort.
- Waste is usually written down after it happens, so nothing guides the next meal.

Existing alternatives: paper logbooks, spreadsheets and the manager's gut feeling.

4 SOLUTION

- Waste forecast in kg for each planned meal, using menu, attendance, event and weather.
- Insights on which meals, menus and events waste the most.
- Simple planner screen: enter tomorrow's plan, see the predicted waste.

8 KEY METRICS

- Forecast error (MAE, kg) on real kitchen data
- Waste cut vs the kitchen's own baseline
- Active kitchens and weekly forecasts used
- Renewals each academic year

3 UNIQUE VALUE PROPOSITION

Know how much food will be wasted before you cook it, and cook just enough.

Saves food cost, cuts waste, and supports UN SDG 12: Responsible Consumption and Production.

5 UNFAIR ADVANTAGE

- Each kitchen's own history trains its model, so it improves with use.
- Built around campus patterns: festivals, exams, mess timings.
- Simple, explainable models managers can trust.

9 CHANNELS

- Direct pilots with campus admin and mess committees
- SDG 12 and sustainability events, college fests
- Web app, nothing to install
- Word of mouth, LinkedIn and a public GitHub demo

2 CUSTOMER SEGMENTS

- College canteens and hostel messes
- Corporate and office cafeterias
- Community kitchens and NGOs

Early adopters: hostel mess committees and campuses that already keep daily meal records.

7 COST STRUCTURE

- Cloud hosting for the model and web app
- Founder development time
- Onboarding each kitchen: data cleanup and setup
- Pilot outreach and travel
- Support and maintenance

6 REVENUE STREAMS

- Monthly subscription per kitchen: [YOUR PRICE, set after pilots]
- Free pilot for first campuses, paid once savings are shown
- Annual licence for campuses with many messes
- Waste reports for SDG 12 sustainability reporting (add-on)

Lifetime value: kitchens renew every academic year if waste and food cost fall.

Status: working prototype trained on 1,800 synthetic meal records. Real-world savings need pilot data from actual kitchens.

THANK YOU

IBMSKILLBUILD

For the learning resources, training and mentorship that built my skills in data science and machine learning.

AICTE

For providing the internship platform and the opportunity to learn through real project work.

MENTORS

For their feedback, support and encouragement throughout the project.

I sincerely thank AICTE and IBM SkillsBuild for giving me the opportunity to take part in this internship. Their programme, resources and guidance helped me turn what I was learning into a working project: Wasteland Pantry, an AI/ML system for food waste prediction and reduction.

ASHISH JAKHMOLA | **INTERNSHIP ID: IBMUEML005287**